HARNESSEVAL

Evaluating Coding Agent Scaffolds Through Ablation-Driven Multi-Dimensional Analysis

A unified framework for decomposing harness quality from base LLM capability

Methods of Learning & Scientific Research
FPT School of Business and Technology (FSB)
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v2 Revised after committee review

RESEARCH TEAM

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OUTLINE

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6. **Modular Harness architecture**
7. **Methodology** — 4 phases
8. **Ablation design** 27x150
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10. **Threats & Risks**
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14. **Contributions** & publishability
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1. PROBLEM: THE EVALUATION PARADOX

The performance of a **coding agent** depends on two factors:

A. Base LLM

Claude, GPT, DeepSeek, Qwen...

Many benchmarks already evaluate this

B. Harness / Scaffold

Tool system, context manager, orchestration...

"The operating system of the agent" [A1]

Evidence: Lou et al. (2026) [A2] show a small model with a good harness can beat a large model with a weak harness, reducing invalid actions by 100%. Bui et al. (2026) [A1]: Adaptive Context Compaction cuts token usage by **54%**.

The paradox: When we report "Agent A scored 70%," we do not know which harness component contributed to that result.

2. RESEARCH GAP: THREE GENUINE REASONS IT EXISTS

We analyzed **49 papers** (2023–2026): no work systematically evaluates the harness itself [S5].

1. Role mismatch

Princeton (SWE-Agent), Anthropic (Claude Code) are developers — incentivized to ship new harnesses, not to compare systems as evaluators.

Analogy: car makers vs Euro NCAP / IIHS

2. Field moving too fast

2023–2026: from a handful of harnesses to dozens. Nobody has paused to meta-evaluate while still racing to build.

3. Only indirect evidence

Lou et al. compared harnesses on game tasks [A2]. Robeyns et al. self-improved a harness but did not measure each component [B10].

Research gap: No framework exists to (a) measure the quality of individual harness components, (b) decompose harness vs base-LLM contribution, and (c) compare harnesses on unified quality dimensions.

2.1. WHAT EXISTS VS WHAT IS MISSING

Aspect	What exists	What is missing
Measuring agent **output**	SWE-bench, SWE-Compass, AgentBoard	—
Tool accuracy in isolation	ToolLLM, AnyTool, Gorilla	Tool use INSIDE a coding harness
Memory in isolation	Mem0, A-MEM (on chatbots)	Memory INSIDE a coding harness
Harness design description	OpenDev, AutoHarness	Measuring harness quality
Cross-backend comparison	—	**Completely unaddressed**
Separating harness vs LLM	—	**Completely unaddressed**

3. AIM + 5 OBJECTIVES

Aim: Build the **first unified evaluation framework** for coding-agent harnesses, enabling decomposition and quantification of each harness component's contribution to agent performance, **independent of the base LLM**.

Obj.	Description	Success criterion
O1	Taxonomy: 3 dimensions, 7 metrics, validated by 5 experts	Fleiss kappa ≥ 0.6
O2	Fork & refactor SWE-Agent into a modular harness	+/- 3% of original SWE-Agent
O3	Ablation: 150 tasks x 27 conditions (3x3x3) + pilot study	7,050 evaluations
O4	Two-way ANOVA decomposing harness vs LLM variance	$p < 0.05$ (Bonferroni)
O5	Release open-source toolkit on GitHub	Docs + CI/CD

v2 Reduced from 5 dimensions → **3**, 80 → **27 conditions**, added pilot study, added two-way ANOVA (committee's question).

4. RESEARCH HYPOTHESES

H1

Tool system has the **largest effect size** (Cohen's d > 0.5) among the three harness components on resolve rate.

H2

Context management has a **medium effect size** (Cohen's d 0.3 – 0.5).

H3

Harness configuration explains **at least 20%** of resolve-rate variance ($eta^2 \geq 0.20$), independent of the LLM backend.

H4

Variance across LLM backends **decreases** when harness quality is higher (statistically significant **interaction effect**).

Neutral framing: No specific numbers predicted ("30–40%"). Only testable directions → avoid confirmation bias. **Supporting or refuting a hypothesis is equally valuable.**

5. TAXONOMY: 3 DIMENSIONS — 7 METRICS

Dimension	Metric	Definition	Measurement
D1 Tool Dispatch	**M1.1** Correct Selection Rate	% tool calls picking a tool from the **acceptable set**	2 annotators, Fleiss kappa
	M1.2 Redundant Call Rate	% calls whose output is not used within the next 3 turns	Automatic log analysis
	M1.3 Utilization Breadth	unique tools used / total available	Automatic log analysis
D2 Context Utilization	**M2.1** Info Retention	BERTScore between full-context and compacted-context output	30 pairs of human preference
	M2.2 Effective Token Ratio	% tokens relevant to the current task	LLM classifier (Haiku) + 200 human labels
D3 Backend Portability	**M3.1** Cross-Backend StdDev	std(resolve rate) across 3 LLM backends	Direct
	M3.2 Min/Max Ratio	min(RR) / max(RR)	Higher = more portable

v2 M1.1: changed "single ground truth" → **acceptable-tool set** (committee Q#4). M2.2: changed vague "token waste" → **LLM classifier + human validation** (committee Q#9).

6. MODULAR HARNESS ARCHITECTURE — 3 SWAPPABLE LAYERS

LAYER 1: Tool Dispatch — `ToolProvider`

Full (12 tools) | Medium (8) | Minimal (5)

LAYER 2: Context Management — `ContextStrategy`

Full History | Sliding Window (50K) | Summary / ACC (2K)

LAYER 3: LLM Backend — `LLMBackend`

Claude Sonnet 4 | GPT-4o | DeepSeek-V3

L4 Safety & Permission — future work

L5 Orchestration — kept unchanged from SWE-Agent

Techniques: Strategy Pattern + Dependency Injection + Factory Pattern. Each layer is swapped via a `YAML config`. Forked from SWE-Agent (MIT, ~15k LOC).

7. METHODOLOGY — FOUR EXPERIMENTAL PHASES

P1
Taxonomy
5 experts validate, Fleiss kappa ≥ 0.6
Weeks 1–6

P1.5
Pilot v2
5 cond x 20 tasks x 2 runs = **200 evals**. ~$80.
Weeks 7–8

P2
Modular Fork
SWE-Agent refactor, verify +/- 3% on 50 tasks
Weeks 9–11

P3
Full Ablation
27 cond x 150 tasks = **7,050 evals**
Weeks 12–18

Why a pilot? Validate that metrics actually discriminate between conditions, estimate cost accurately, debug the pipeline, and compute preliminary effect sizes for power analysis. **Prevents blind commitment** to the expensive full experiment.

8. ABLATION DESIGN: 27 CONDITIONS = 3 X 3 X 3

27	150	7,050	~$2.5k
CONDITIONS	TASKS	EVALUATIONS	BUDGET

Factor	Level 1	Level 2	Level 3
Tool Config	Full (12)	Medium (8)	Minimal (5)
Context Strategy	Full History	Sliding Window 50K	Summary / ACC 2K
LLM Backend	Claude Sonnet 4	GPT-4o	DeepSeek-V3

Dataset: 150 tasks from SWE-bench Verified, stratified random sampling by difficulty. **Runs:** 10 critical conditions x 3 runs + 17 conditions x 1 run. **Power analysis:** alpha = 0.05, power ≥ 0.80.

9. TWO-WAY ANOVA — DECOMPOSING HARNESS VS LLM

"Independent of the LLM" does **not** mean harness and LLM are uncorrelated. It means we can decompose the variance due to harness vs the variance due to LLM.

Statistical model:

```
Total Variance = Var(Harness) + Var(LLM) + Var(Harness x LLM) + Error

Factor A: Harness config   (9 levels: 3 tools x 3 contexts)
Factor B: LLM backend      (3 levels: Claude, GPT, DeepSeek)
DV:        Resolve rate

If Var(Harness) is significant    --> harness has an INDEPENDENT effect
If Var(H x LLM) is significant    --> there is an INTERACTION
eta-squared                       --> % of variance explained per factor
```

Main theoretical contribution: Formalizing how to separate harness quality from LLM quality. Post-hoc: Tukey HSD pairwise, Cohen's d effect sizes, Bonferroni correction for 7 metrics.

10. THREATS TO VALIDITY & RISK ANALYSIS

THREATS

Type	Mitigation
Internal: tool-removal confound	Reduce 12→8→5; log which tools each task needs
Internal: LLM non-determinism	Temperature = 0, 3 runs for critical conds
External: Python only	Caution when generalizing
Construct: M1.1 annotator-dependent	2 annotators, Fleiss kappa

TOP RISKS

Risk	P	Mitigation
API cost overrun	High	Pilot estimates, DeepSeek cheap
Docker env instability	Med	Pre-filter, retry x2
Metrics not discriminative	Med	Detected early via pilot
DeepSeek API unstable	Low	Backup: Qwen2.5-72B

v2 V1 had no Threats or Risk Analysis — the committee explicitly requested this.

11. TIMELINE (24 WEEKS) & BUDGET

Phase	Weeks	Content	Deliverable
P1	1–6	Taxonomy + expert validation	Taxonomy v1.0, kappa report
P1.5	7–8	**Pilot study** (200 evals)	Metrics validation, cost estimate
P2	9–11	Fork SWE-Agent + modular refactor	Modular harness v1.0
P3	12–18	Full ablation (7,050 evals)	Raw data, ANOVA results
P4	19–22	Analysis + guidelines + thesis writing	Draft thesis
P5	23–24	Review, revisions, toolkit release	Final + GitHub release

Estimated budget

Pilot: ~$80
Full experiment: ~$2,500
Contingency: ~$500
Total: $2,500 – $3,100

v1 vs v2 comparison

v1: $800–1,200 (unrealistic)
v2: $2,500–3,100 (includes retry + pilot)
Evaluations: 16,000 → 7,050 (-56%)

12. IMPLEMENTATION STATUS — ALREADY BUILT

167	9	27	360
TESTS PASSING	PHASES DONE	CONDITIONS GENERATED	TRAJECTORIES COLLECTED

TOOLKIT `harness_eval/`

- `configs/` — 27 conditions (3x3x3)
- `metrics/` — 7 metrics M1.1–M3.2
- `parsers/trajectory.py` — SWE-Agent log parser
- `pipeline/analysis.py` — ANOVA, Cohen's d, GLMM
- `pipeline/runner.py` — Dry-run + SWE-Agent + Ollama
- `harness/` — 3 ABCs + 9 providers + factory
- `cli.py` — info / pilot / run / analyze / convert

STREAMLIT DASHBOARD

- 5 tabs: Config / Run Monitor / Logs / Compare / ANOVA
- Run modes: **ollama** / **dry-run** / **real**
- Parallel ThreadPoolExecutor (4 workers)
- 150 SWE-bench tasks embedded (difficulty-aware)
- Ollama: qwen2.5:7b, deepseek-r1:1.5b
- One-click setup: `start_harness_eval.sh`

13. PRELIMINARY RESULTS — EMPIRICAL ANOVA

Run on the current **trajectories/** folder: 360 observations, 18 conditions (2 backends: claude + sonnet)

Source	eta^2	F	p-value	Sig?
Tool	0.0539	10.264	**< 0.001**	***
Context	0.0248	4.723	**0.0095**	**
Backend	0.0025	0.959	0.3282	—
Tool x Context	0.0058	0.550	0.6988	—
Tool x Backend	0.0069	1.314	0.2701	—
Context x Backend	0.0069	1.314	0.2701	—
Error	0.8979	—	—	—

Preliminary observations:

- H1 direction **Tool** has the largest effect (eta^2 = 0.054 > Context = 0.025)
- H2 direction **Context** has a medium, statistically significant effect
- Need more data Backend has only 2 levels (haiku empty); H4 requires the full design

14. CONTRIBUTIONS & PUBLISHABILITY

Theoretical
Formalizing how to separate **harness quality from LLM quality** via two-way ANOVA.

Empirical
First quantitative decomposition of each harness component (Tool / Context / Backend).

Toolkit
Reusable evaluation framework — open-source on GitHub, docs, CI/CD.

PRACTICAL IMPACT

- **Developers:** know which harness component to prioritize for improvement
- **Researchers:** standard framework to compare new harnesses
- **Industry:** data-driven guidelines for harness design

PUBLISHABILITY

Venue	Type	Fit
ICSE 2027 NIER	Conference (New Ideas)	Strong fit
MSR 2027	Conference	Fit
LLM Agents Workshop (NeurIPS/ICML)	Workshop	Fit

Q&A

Thank you for your attention
and thoughtful questions

Resources:

`NCKK-Docs/de-tai/de-cuong/DE-CUONG-HarnessEval-v2.md`
`NCKK-Docs/de-tai/harness-eval/` — toolkit + 167 tests
`plans/Plan-Coding-HarnessEval.md` — session log